

B.E. ELECTRICAL ENGINEERING SECOND YEAR SECOND SEMESTER - 2025**SUBJECT: FIELD THEORY**

Time: Three Hours

Full Marks: 100
(50 Marks for each part)**Use a separate Answer-Script for each part**

Two marks are reserved for neat and well-organized answers

Question No.	Part-I	Marks
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Answer any three questions

1. (a) Derive a correlation between rise of electric flux with the charge per unit volume. 5
- (b) Derive the governing equation which is used to evaluate the potential at a point within a charge free dielectric medium. 4
- (c) The potential in a space containing a dielectric medium of ϵ_{r1} is given by $\phi_1 = -7xy + 3yz - 4zx$ V, and another potential in a space containing a dielectric medium of ϵ_{r2} is given by $\phi_2 = -2x + 7y - 5yz$ V, where x, y and z are in meters. If the y -component of electric flux density at the point (1,2,2) m is same for both the cases, then find the ratio of ϵ_{r1} and ϵ_{r2} . 7
2. (a) A disc charge having radius of R and charge density of σ C/m² is placed along y -axis with its center at origin. A ring charge having radius of R and charge density of λ C/m is placed along x -axis with its center at (2R,0,0). Determine the electric field at a point having co-ordinate of (4R, 0, 0) with proper derivations. 7
- (b) Express the Poisson's equation in cylindrical coordinate system. 6
- (c) Two concentric spherical electrodes of radii 10m and 20m, respectively, are having their centers at (0,0,100)m. The inter-spherical volume is filled with air. The inner sphere is positively charged and the outer sphere is at zero potential. If a unit positive charge is located at (0,21.21,121.21)m. Explain in which direction the unit charge will move. 3
3. (a) Discuss how the volume density of energy storage within parallel plate arrangement is changed if the area of the plates is doubled and the distance between the plates is halved. 8
- (b) How do the electric field quantities vary if the electric flux lines pass through a multi-dielectric medium? 8

[Turn over

4. (a) Justify or correct the following statement giving reasons: "Electric field intensity just off the conductor surface can be greater than the electric field intensity on the conductor surface". 6
- (b) Discuss the reasons behind the development of mechanical pressure on the conductor surface. Also, derive expression of the mechanical pressure. 3
- (c) Justify with reasons whether two different solutions are possible for the electric potential function within a volume V bounded by a surface S , with a given boundary condition on the surface S , when the volume V contains no free charge within it. 7
5. (a) How does the capacitance of infinite two wire transmission line depend on their radius and separation distance? Justify through proper derivation. 8
- (b) Correct or justify the following statement with reasons: "The relative permittivity of the liquid dielectric used for impregnation must be less than half of the relative permittivity of the solid insulation being impregnated". 8

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PART-II

Answer any **three** questions. Two marks are for neatness. All symbols have their usual significance.

1. a) What is the fundamental definition of curl of a vector field? 8+8=16
Derive the expression for curl of a vector field $\mathbf{M}$, written as $\nabla \times \mathbf{M}$ where, ∇ is the vector differential operator in Cartesian co-ordinates.
- b) Establish the Stoke's Theorem and $\nabla \times \mathbf{H} = \mathbf{J}$.
2. a) Derive the expression for Divergence of a vector field $\mathbf{B}$ in Cartesian co-ordinates using fundamental definition. 6+5+5=16
b) Explain the physical significance of $\nabla \cdot \mathbf{J} = 0$.
c) Explain that $\nabla \cdot \mathbf{B} = 0$
3. a) Establish $\nabla \times \mathbf{H} = \mathbf{J} + \partial \mathbf{D} / \partial t$ and show the pictorial view of this relation. 8+8=16
b) Derive $\nabla \times \mathbf{E} = - \partial \mathbf{B} / \partial t$ and show the pictorial view of this relation.
4. a) Derive Electromagnetic Wave equations from Maxwell's equations. 6+10=16
b) Obtain E.M. wave equations in free space and then find an analytical solution of the wave equation of Magnetic Field ($\mathbf{H}$) in free space considering it as a plane wave and also draw the wave propagation.
5. Write short notes on any four: 4×4=16
 - a) Physical significance of Curl;
 - b) Poynting Vector;
 - c) EM Plane wave velocity in free space;
 - d) Displacement current;
 - e) Divergence of the curl of any vector field is always zero.